

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PURWANCHAL CAMPUS

A MINOR PROJECT REPORT
ON
A LIGHTWEIGHT DUAL AGGREGATION TRANSFORMER FOR IMAGE SUPER-RESOLUTION

BY

MILAN ARJEL (PUR078BCT050)

SAMIR BIDARI (PUR078BCT073)

SANDEEP POUDEL (PUR078BCT075)

SANDESH POKHAREL (PUR078BCT076)

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

PURWANCHAL CAMPUS

DHARAN, NEPAL

2 March, 2025

A LIGHTWEIGHT DUAL AGGREGATION TRANSFORMER FOR IMAGE SUPER-RESOLUTION

By

Milan Arjel (PUR078BCT050)

Samir Bidari (PUR078BCT073)

Sandeep Poudel (PUR078BCT075)

Sandesh Pokharel (PUR078BCT076)

Project Supervisor

Asst. Prof. Pukar Karki

A project submitted to the Department of Electronics and Computer Engineering in partial fulfillment of the requirements for the Bachelor's Degree in Computer Engineering

Department of Electronics and Computer Engineering
Purwanchal Campus, Institute of Engineering
Tribhuvan University
Dharan, Nepal

March , 2025

ABSTRACT

Image Super-Resolution (ISR) aims to reconstruct high-resolution (HR) images from low-resolution (LR) images. However, traditional CNN-based models cannot provide the desired performance, while transformer-based models are computationally expensive. To address this challenge, we present a Lightweight Dual Aggregation Transformer, a lightweight yet effective transformer-based model for ISR.

Lightweight Dual Aggregation Transformer (LDAT) uses a smart framework that combines two key components to process images effectively. First, it has a part called Dual Spatial Transformer Blocks (DSTB), which focuses on understanding the spatial details—like how different parts of an image relate to each other—using a flexible attention system and a sliding window technique. Second, it has Dual Channel Transformer Blocks (DCTB), which looks at the relationships between different color channels in the image, again using an adaptable attention approach. Together, these ensure the system captures all the important features. There's also a Spatial-Guided Feedforward Network (SGFN) that polishes these features further. Finally, a simple and efficient method called pixel shuffle is used to create a high-resolution image without needing too much computing power.

Tests on standard datasets show that LDAT offers competitive performance as other methods while using fewer resources and being less complicated. This makes it great for applications that need to work quickly or on small devices.

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to our supervisor Asst. Prof. Pukar Karki, the Deputy Head of the Department, for their invaluable guidance, insightful suggestions, and unwavering support throughout the course of this project. Their constructive feedback and expertise have been instrumental in shaping this work.

We extend our sincere thanks to our esteemed teachers and faculty members for their encouragement, valuable feedback, and scholarly insights. Their dedication to academic excellence and their commitment to nurturing knowledge have been a constant source of inspiration. Their collective support has played a significant role in refining and enhancing the quality of this project.

MILAN ARJEL (PUR078BCT050)
SAMIR BIDARI (PUR078BCT073)
SANDEEP POUDEL (PUR078BCT075)
SANDESH POKHAREL (PUR078BCT076)

TABLE OF CONTENTS

ABSTRACT i

ACKNOWLEDGEMENT ii

LIST OF FIGURES v

LIST OF TABLES vi

LIST OF ABBREVIATIONS viii

1 INTRODUCTION 1

1.1 Background 1

1.2 Gap Identification 1

1.3 Motivation 1

1.4 Objectives 1

2 LITERATURE REVIEW 2

3 METHODOLOGY 4

3.1 System Design and Block Diagrams 4

3.1.1 Model Architecture 4

3.1.2 Dual Aggregation Transformer Block 4

3.1.3 Dual Feature Aggregation 5

3.2 Algorithm 6

3.3 Software Development Model 6

4 EXPERIMENTS 8

4.1 Implementation 8

4.2 Dataset Preparation 8

4.3 Training 8

4.4 Training Curves 9

5 RESULTS **11**

5.1 Visual Comparisons 11

5.2 Quantitative Evaluation 12

6 CONCLUSION **14**

REFERENCES **15**

LIST OF FIGURES

Figure 3.1: The network architecture of Lightweight Dual Aggregation Transformer (LDAT). 4

Figure 3.2: Dual Aggregation Transformer Block (DATB). 5

Figure 3.3: Agile Model 7

Figure 4.1: Loss curve over 360 epochs. The decreasing loss indicates stable convergence. 9

Figure 4.2: PSNR curve over the 360 epochs. The increasing trend shows improved image reconstruction quality. 9

Figure 4.3: SSIM curve over the 360 epochs. Higher SSIM values indicate better structural similarity preservation. 10

Figure 5.1: Visual comparison of super-resolution results for an image from the Urban100 dataset. The left image represents the reconstructed high-resolution result, while the right grid highlights zoomed-in regions. . . . 11

Figure 5.2: Visual comparison of super-resolution results for an image from the Set14 dataset. The left image represents the reconstructed high-resolution result, while the right grid highlights zoomed-in regions. . . . 12

LIST OF TABLES

Table 5.1: Performance metrics (PSNR/SSIM) for $\times 4$ super-resolution on different datasets. 12

Table 5.2: Details about the dataset 13

Table 5.3: Performance Accuracy for Baseline Models 13

LIST OF ABBREVIATIONS

DAT	: Dual Aggregation Transformer
LDAT	: Lightweight Dual Aggregation Transformer
DATB	: Dual Aggregation Transformer Block
DSTB	: Dual Spatial Transformer Block
DCTB	: Dual Channel Transformer Block
AIM	: Adaptive Interaction Module
SW-SA	: Spatial Window Self-Attention
CW-SA	: Channel-Wise Self-Attention
FFN	: Feed-Forward Network
SGFN	: Spatial-Gate Feed-Forward Network
DW-Conv	: Depth-Wise Convolution
RCAN	: Residual Channel Attention Network
SR	: Super-Resolution
ISR	: Image Super-Resolution
HR	: High-Resolution
LR	: Low-Resolution
GAN	: Generative Adversarial Network
ESRGAN	: Enhanced Super Resolution Generative Adversarial Network
GPU	: Graphhics Processing Unit
PSNR	: Peak Signal-to-Noise Ratio
SSIM	: Structural Similarity Index
FLOP	: Floating-point Operation
SE	: Squeeze-and-Excitation
ECA	: Efficient Channel Attention
MSE	: Mean Squared Error
MAE	: Mean Absolute Error
ESPCN	: Efficient Sub-pixel Convolutional Neural Network
SRCNN	: Super-Resolution Convolutional Neural Network
FSRCNN	: Fast Super-Resolution Convolutional Neural Network
CARN	: Convolutional Anchored Regression Network
FMEM	: Fast and Memory Efficinet Network

CHAPTER 1

INTRODUCTION

1.1 Background

Image super-resolution (SR)[1] is the process of reconstructing high-resolution (HR) images from low-resolution (LR) inputs. It plays a crucial role in various fields, such as medical diagnostics[2], remote sensing [3], and multimedia enhancement. Recent advances in SR include models such as SwinIR, a transformer-based architecture, and ESRGAN, which utilizes generative adversarial networks (GANs). These models have demonstrated exceptional performance, setting benchmarks for high-fidelity image reconstruction.

1.2 Gap Identification

Despite their impressive results, state-of-the-art models such as SwinIR [4] and ESRGAN [5]have significant drawbacks: their large model sizes and high computational demands. These limitations make them unsuitable for deployment in real-world scenarios. This creates a gap in lightweight SR models that can provide good performance while also being more efficient than the state-of-the-art models.

1.3 Motivation

The growing need for SR solutions in mobile and edge applications shows the importance of having efficient and lightweight models. Although current models perform well, they require too many resources to be used in real-world situations. This drives the need for a simpler, yet effective model that balances performance and efficiency.

1.4 Objectives

- To develop a lightweight version of the Dual Aggregation Transformer (LDAT) that achieves high-quality image super-resolution while reducing computational costs and memory requirements.

CHAPTER 2

LITERATURE REVIEW

Image super-resolution (SR) aims to reconstruct high-resolution (HR) images from low-resolution (LR) inputs, with significant advancements made using deep learning techniques. Traditional methods, such as interpolation-based approaches (e.g., bicubic interpolation), offered limited quality improvements due to their inability to recover high-frequency details. Compared to traditional image restoration methods which are generally model-based, learning based methods, especially CNN-based methods, have become more popular due to their impressive performance.They often learn mappings between low-quality and high-quality images from large-scale paired datasets. SRCNN [6]is the pioneering work, which first utilizes CNN and outperforms traditional approaches.To build more effective models for SR, the recently developed methods tend to employ deeper and more complicated architectures as well as the attention techniques.An enhanced residual block was proposed in[7] to train the deep model without batch normalization. Zhang et al. proposed a residual-in-residual structure coupled with channel attention to train a very deep network over 400 layers. Besides, attention mechanisms [8–11]in terms of spatial or channel dimensions are adopted to achieve further improvement in modeling ability. Al-though these models achieved competitive results, they are pure CNN-based model. This means that they can only extract local features and cannot learn the global information,which is not conducive to the restoration of texture details.

The breakthrough of transformer networks in natural language processing (NLP) inspired of use of self-attention (SA) in computer vision tasks. The SA mechanism in transformers can effectively model the dependency across data, and it has achieved impressive results on several high-level vision tasks, such as image classification [12, 13], image detection[14, 15], and segmentation[16]. To produce the sequence elements, ViT flattened the 2D image patches in a vector and fed them into the Transformer.A series of Transformer-based methods are proposed to improve the efficiency and effectiveness of Transformer for high-level tasks. Very recently, transformer has also been applied to low-level vision tasks[17] . IPT[18] is an extremely large pre-trained model

for various low-level vision tasks based on the standard vision transformer. It computes both the local feature and SA on non-overlapped patches, which however may lose some useful information for reproducing image details. SwinIR hence adapts the Swin Transformer [19]to image restoration, which combines the advantages of both CNNs and transformers. Swin Transformer applies local windows to limit the attention scope and shift operations to increase the window interaction . Though SwinIR has achieved impressive results for image SR, its network structure is mostly borrowed from the Swin Transformer, which is designed for high-level vision tasks. In particular, the network design of SwinIR is redundant for the SR problem, and it calculates SA on small fixed-size windows, preventing it from exploiting long-range feature dependency. Restormer [20] operates self-attention along channel dimensions and applies the UNet architecture[21] . These methods remarkably outperform CNN-based methods. It reveals that both spatial and channel information are important for performance.Although these methods achieved promising results, they require a lot of training data and need heavy GPU memory to train the model .

CHAPTER 3

METHODOLOGY

This methodology details the approach to designing the lightweight DUAL AGGREGATION TRANSFORMER (LDAT) tailored for efficient image SR, both performance and computational efficiency. The focus is on achieving performance comparable to that of leading models, while reducing computational cost and memory demands.

3.1 System Design and Block Diagrams

3.1.1 Model Architecture

The DAT network consists of three main modules: shallow feature extraction, deep feature extraction, and image reconstruction. First, a convolution layer processes the low-resolution (LR) input to generate shallow features. These features are then passed through the deep feature extraction module, which consists of multiple residual groups (RGs). Each RG contains dual aggregation transformer blocks (DATBs) that apply spatial and channel self-attention, with residual connections ensuring training stability. Finally, the high-resolution (HR) output is reconstructed using the pixel shuffle method and convolution layers to refine features before and after upsampling.

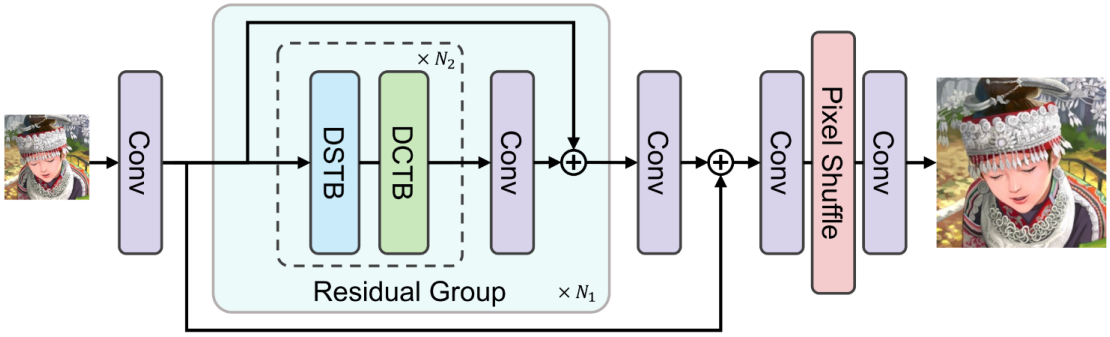


Figure 3.1: The network architecture of Lightweight Dual Aggregation Transformer (LDAT).

3.1.2 Dual Aggregation Transformer Block

The Dual Aggregation Transformer Block (DATB) is the core component of the proposed method and consists of two types: Dual Spatial Transformer Block (DSTB)

and Dual Channel Transformer Block (DCTB). DSTB employs spatial window self-attention (SW-SA) to capture spatial dependencies, while DCTB utilizes channel-wise self-attention (CW-SA) to model global channel relationships. By alternating DSTB and DCTB, the model effectively aggregates spatial and channel features across layers.

To enhance feature interaction, the Adaptive Interaction Module (AIM) fuses global self-attention and local convolution features, introducing adaptive spatial self-attention (AS-SA) and adaptive channel self-attention (AC-SA). This adaptive mechanism ensures better coupling of local and global information, improving feature representation.

Additionally, the Spatial-Gate Feed-Forward Network (SGFN) enhances feature modeling by integrating a spatial gating mechanism using depth-wise convolutions, allowing for better spatial information capture while reducing channel redundancy.

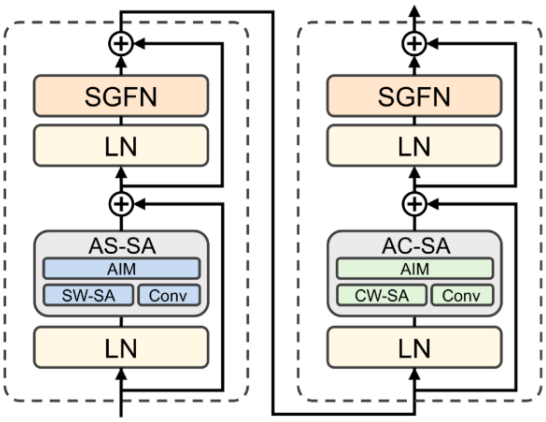


Figure 3.2: Dual Aggregation Transformer Block (DATB).

3.1.3 Dual Feature Aggregation

LDAT achieves dual feature aggregation at two levels:

- **Inter-block aggregation**
  - Alternates **DSTB** and **DCTB** for complementary learning.
  - Facilitates efficient feature flow between consecutive blocks.
  - Enhances both **spatial and channel feature learning**.
- **Intrablock aggregation**
  - **AIM** and **SGFN** refine spatial and channel interactions within each block.

  - Strengthens both **local and global feature representations**.

- **Overall impact**
  - Captures both **spatial and channel dependencies** effectively.
  - Improves **feature extraction and representation**.
  - Enhances performance in **high-resolution image reconstruction** tasks.

3.2 Algorithm

Algorithm 1 Training a Neural Network

Initialize model parameters θ.
for epoch $e = 1$ to E **do**
    Shuffle training data D.
    Divide D into batches of size B.
    for each batch B_j **do**
        Compute predictions: $\hat{y} = f_\theta(x)$.
        Compute loss: $\mathscr{L}(\hat{y}, y)$.
        Compute gradients: $\nabla_\theta \mathscr{L}$.
        Update parameters: $\theta = \theta - \alpha \cdot \nabla_\theta \mathscr{L}$.
    end for
    Compute validation loss.
end for
return optimized parameters θ^*.

3.3 Software Development Model

The development of the lightweight Dual Aggregation Transformer (LDAT) follows the Agile model, an iterative and incremental approach that emphasizes flexibility, collaboration, and continuous improvement. The process begins with an initial planning phase to outline high-level requirements and goals, followed by a series of short development cycles, or sprints. Each sprint involves requirement refinement, system design, implementation, testing, and evaluation, allowing the team to adapt to evolving needs and incorporate feedback throughout the development lifecycle. This iterative approach ensures that the DAT model is progressively refined and aligned with user expectations.

For the implementation phase, a web-based interface is developed to facilitate user interaction with the DAT model. The backend is built using FastAPI, a high-performance Python framework, which provides an efficient and scalable API for processing image

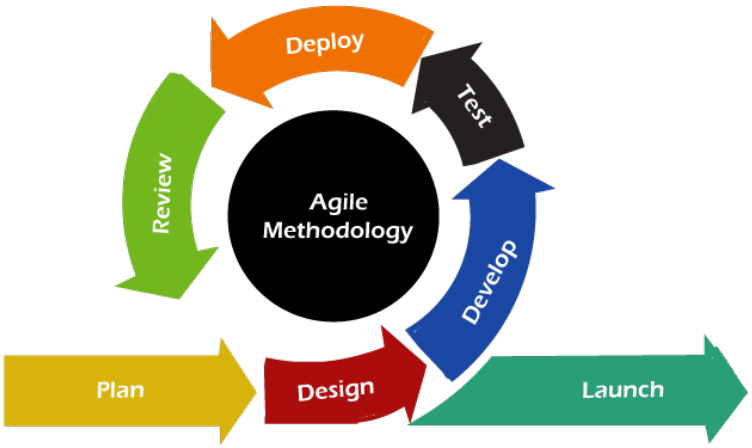


Figure 3.3: Agile Model

super-resolution requests. FastAPI's asynchronous capabilities enable rapid handling of low-resolution image inputs, delivering high-resolution outputs with minimal latency. The front-end is crafted using React, a JavaScript library for creating dynamic and responsive user interfaces. React supports the development of an intuitive interface where users can upload images, tweak parameters (e.g., scaling factor), and view super-resolved results in real-time. Throughout the Agile sprints, the integration of FastAPI and React is achieved via RESTful API endpoints, ensuring smooth communication between the back-end model inference and the front-end display. This combination, refined iteratively based on user feedback and testing, delivers a robust, adaptable, and user-friendly platform for deploying and interacting with the DAT model.

CHAPTER 4

EXPERIMENTS

4.1 Implementation

The lightweight Dual Aggregation Transformer (LDAT) is built with 4 residual groups (RGs), and each RG contains 2 pairs of dual aggregation transformer blocks (DATBs). Each DATB contains 1 dual spatial transformer block (DSTB) and 1 dual channel transformer block (DCTB). The attention head number, channel dimension, and channel expansion factor in the Spatial-Gate Feed-Forward Network (SGFN) are set to 4, 60, and 3 for both DSTB and DCTB. For all DSTBs, we set the window size to 8×16. The image range of image feed to the model is set to 0-255.

4.2 Dataset Preparation

We adopt the high-quality DF2K dataset (a combination of DIV2K [22] and Flickr2K [23]) as our training dataset. DF2K consists of 3,450 high-quality training images, combining 800 images from DIV2K and 2,650 images from Flickr2K, along with 100 validation images. For validation purposes, we use the first 10 images from the DF2K validation dataset to reduce training time per epoch. For testing, we use the Set5 [24], Set14 [25], and Urban100 [26] datasets.

4.3 Training

LDAT is trained with a patch size of 64×64 and a batch size of 16 for 215 epochs and increased batch size to 32 for further training upto 360 epochs. Optimization is performed using the Adam optimizer by minimizing the L1 loss. The initial learning rate is set to 5e-4 and is halved at [80,130,160,180, 250, 260, 300] epochs. Data augmentation includes random rotations (90°, 180°, 270°) and horizontal flips. The model is implemented in PyTorch and trained using 2 T4 GPUs.

4.4 Training Curves

To analyze the convergence and performance of our model, we present the loss curve, PSNR curve, and SSIM curve during training. These plots help illustrate how the model improves over time.

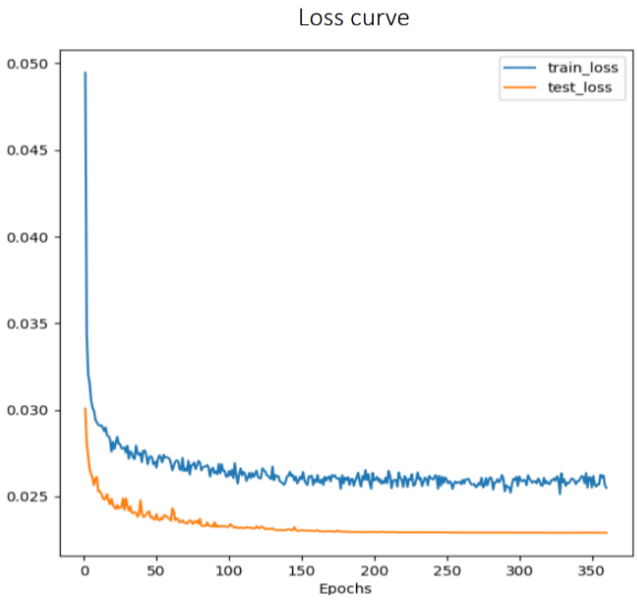


Figure 4.1: Loss curve over 360 epochs. The decreasing loss indicates stable convergence.

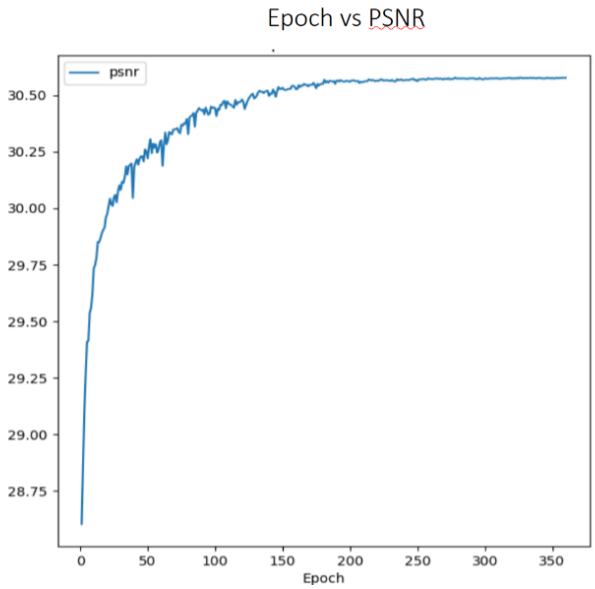


Figure 4.2: PSNR curve over the 360 epochs. The increasing trend shows improved image reconstruction quality.

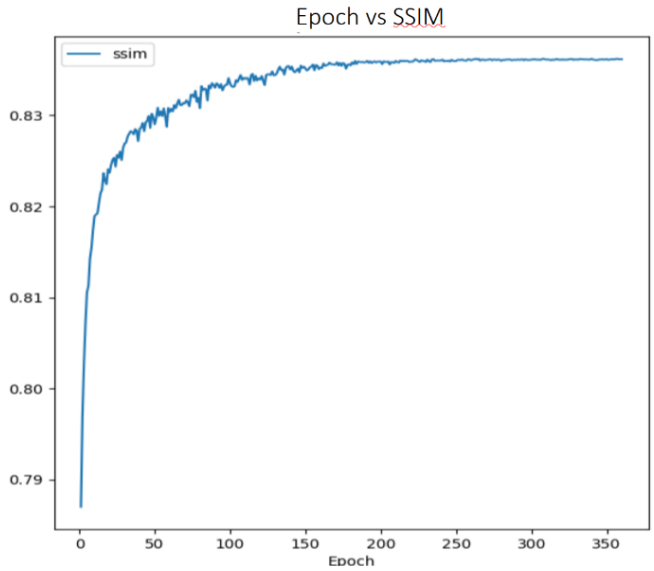


Figure 4.3: SSIM curve over the 360 epochs. Higher SSIM values indicate better structural similarity preservation.

CHAPTER 5

RESULTS

The experimental results highlight the effectiveness of the proposed method across multiple benchmark datasets. As shown in Table 5.1, our approach achieves competitive PSNR and SSIM values compared to other lightweight super-resolution models, particularly on Set5 and Set14. While performance variations exist across datasets, the overall results demonstrate the robustness of our model in reconstructing fine details and preserving structural integrity.

To further illustrate the qualitative performance, Figures 5.1 and 5.2 present visual comparisons of super-resolved images from the Urban100 and Set14 datasets. Each figure includes a high-resolution reconstruction alongside zoomed-in regions, allowing for a detailed assessment of texture fidelity and edge preservation. The proposed method effectively enhances image quality, reducing artifacts while maintaining sharpness and clarity.

5.1 Visual Comparisons

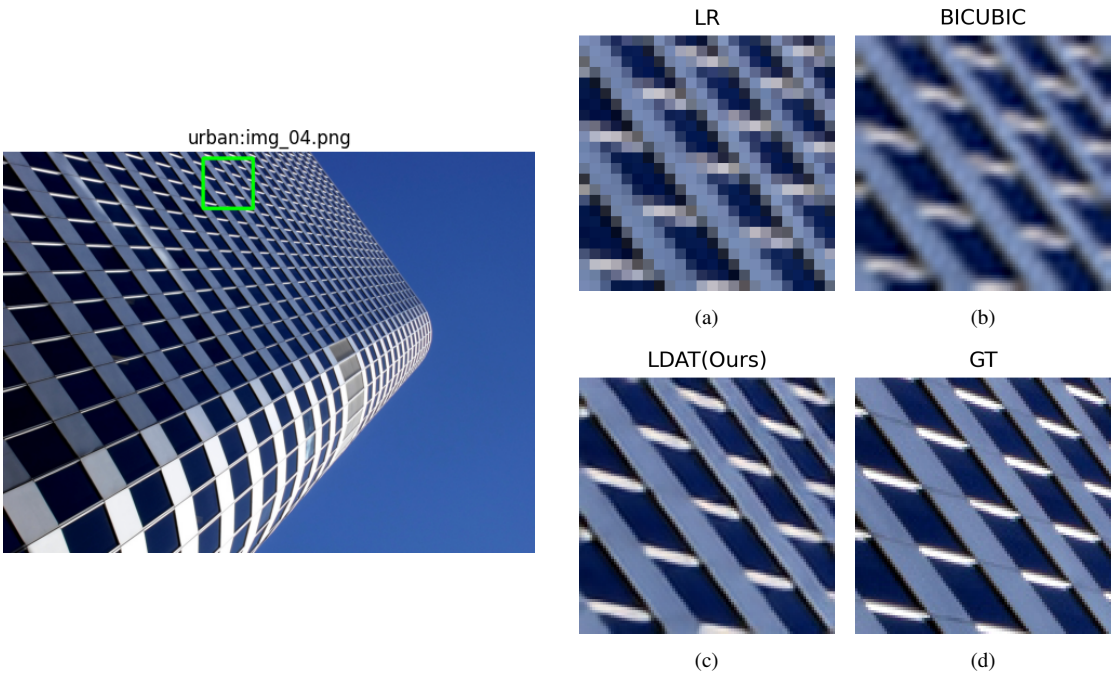


Figure 5.1: Visual comparison of super-resolution results for an image from the Urban100 dataset. The left image represents the reconstructed high-resolution result, while the right grid highlights zoomed-in regions.



Figure 5.2: Visual comparison of super-resolution results for an image from the Set14 dataset. The left image represents the reconstructed high-resolution result, while the right grid highlights zoomed-in regions.

5.2 Quantitative Evaluation

The quantitative results in Table 5.1 compare the performance of our model against state-of-the-art methods across multiple datasets. Our approach delivers competitive results, demonstrating its ability to generate high-fidelity images with well-preserved textures and minimal artifacts.

Method	**Parameters(K)**	**PSNR / SSIM**		
		Set5	**Set14**	**Urban100**
Bicubic	-	28.42 / 0.8104	26.00 / 0.7027	23.14 / 0.6577
ESRT[27]	751	32.19 / 0.8947	28.69 / 0.7833	26.39 / 0.7962
SwinIR-light[4]	897	32.44 / 0.8976	28.77 / 0.7858	26.47 / 0.7980
ELAN-light[28]	601	32.43 / 0.8975	28.78 / 0.7858	26.54 / 0.7982
LDAT (Ours)	738	32.37 / 0.8940	28.63 / 0.7825	26.09 / 0.7860

Table 5.1: Performance metrics (PSNR/SSIM) for $\times 4$ super-resolution on different datasets.

Table 5.2: Details about the dataset

	Human Writing	**AI generated**
Size	3388	3388
Avg. Length	323	301
Max Length	13112	1000
Min Length	200	47

Table 5.3: Performance Accuracy for Baseline Models

Baseline Models	**Accuracy**		
	64 tokens	**128 tokens**	**256 tokens**
ViT5-base	0.8611	0.8438	0.8591
BARTpho-syllable-base	0.8570	0.8840	0.8632
PhoBERT-base	0.8743	0.8819	0.8708
mDeBERTa-v3-base	0.8957	0.8542	0.7975
BERT-multilingual-base-cased	0.8017	0.8964	0.8279
Avg	**0.8580**	**0.8721**	**0.8437**

CHAPTER 6

CONCLUSION

In this paper, we propose a Lightweight Double Aggregation Transformer (LDAT). This model is composed of three parts : shallow feature extraction , deep feature extraction and HR reconstruction modules . It aggregates both Spatial and Channel information which is important for the performance of Super-Resolution models.For this our model alternatively applies spatial window and channel-wise self-attention on each transformer block.DAT can model global dependencies through this alternate strategy and achieve inter-block feature aggregation among spatial and channel dimensions.Adaptive interaction module (AIM) and the spatial-gate feed-forward network (SGFN) were used to enhance each block and realize intra-block feature aggregation between two dimensions.Although our model achieves comparable performance to other lightweight transformer based models but still it is computation and memory-intensive comapred to those lightweight CNN-based models .In the future, we will further explore more efficient implementations for more low-level vision tasks.

REFERENCES

[1] Z. Wang, J. Chen, and S. C. Hoi, "Deep learning for image super-resolution: A survey," *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 10, pp. 3365–3387, 2020.

[2] S. Umirzakova, S. Ahmad, L. U. Khan, and T. Whangbo, "Medical image super-resolution for smart healthcare applications: A comprehensive survey," *Information Fusion*, vol. 103, p. 102075, 2024.

[3] K. Bajpai and R. Soni, "Analysis of image enhancement techniques used in remote sensing satellite imagery," *International Journal of Computer Applications*, vol. 169, no. 10, pp. 1–11, 2017.

[4] J. Liang, J. Cao, G. Sun, K. Zhang, L. V. Gool, and R. Timofte, "Swinir: Image restoration using swin transformer," 2021. [Online]. Available: https://arxiv.org/abs/2108.10257

[5] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, C. Dong, C. C. Loy, Y. Qiao, and X. Tang, "Esrgan: Enhanced super-resolution generative adversarial networks," 2018. [Online]. Available: https://arxiv.org/abs/1809.00219

[6] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part IV 13*. Springer, 2014, pp. 184–199.

[7] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 286–301.

[8] Y. Zhang, K. Li, K. Li, B. Zhong, and Y. Fu, "Residual non-local attention networks for image restoration," *arXiv preprint arXiv:1903.10082*, 2019.

[9] B. Niu, W. Wen, W. Ren, X. Zhang, L. Yang, S. Wang, K. Zhang, X. Cao, and H. Shen, "Single image super-resolution via a holistic attention network," in *Com-*

puter Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XII 16. Springer, 2020, pp. 191–207.

[10] Y. Mei, Y. Fan, Y. Zhou, L. Huang, T. S. Huang, and H. Shi, "Image super-resolution with cross-scale non-local attention and exhaustive self-exemplars mining," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 5690–5699.

[11] H. Chen, J. Gu, and Z. Zhang, "Attention in attention network for image super-resolution," *arXiv preprint arXiv:2104.09497*, 2021.

[12] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[13] Y. Li, K. Zhang, J. Cao, R. Timofte, and L. Van Gool, "Localvit: Bringing locality to vision transformers," *arXiv preprint arXiv:2104.05707*, 2021.

[14] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European conference on computer vision*. Springer, 2020, pp. 213–229.

[15] L. Liu, W. Ouyang, X. Wang, P. Fieguth, J. Chen, X. Liu, and M. Pietikäinen, "Deep learning for generic object detection: A survey," *International journal of computer vision*, vol. 128, pp. 261–318, 2020.

[16] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-unet: Unet-like pure transformer for medical image segmentation," in *European conference on computer vision*. Springer, 2022, pp. 205–218.

[17] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "Swinir: Image restoration using swin transformer," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 1833–1844.

[18] H. Chen, Y. Wang, T. Guo, C. Xu, Y. Deng, Z. Liu, S. Ma, C. Xu, C. Xu, and W. Gao, "Pre-trained image processing transformer," in *Proceedings of the*

IEEE/CVF conference on computer vision and pattern recognition, 2021, pp. 12 299–12 310.

[19] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10 012–10 022.

[20] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 5728–5739.

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18*. Springer, 2015, pp. 234–241.

[22] R. Timofte, E. Agustsson, L. Van Gool, M.-H. Yang, and L. Zhang, "Ntire 2017 challenge on single image super-resolution: Methods and results," in *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, 2017, pp. 114–125.

[23] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, July 2017.

[24] M. Bevilacqua, A. Roumy, C. Guillemot, and M. L. Alberi-Morel, "Low-complexity single-image super-resolution based on nonnegative neighbor embedding," 2012.

[25] R. Zeyde, M. Elad, and M. Protter, "On single image scale-up using sparse-representations," in *Curves and Surfaces: 7th International Conference, Avignon, France, June 24-30, 2010, Revised Selected Papers 7*. Springer, 2012, pp. 711–730.

[26] J.-B. Huang, A. Singh, and N. Ahuja, "Single image super-resolution from transformed self-exemplars," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 5197–5206.

[27] Z. Lu, J. Li, H. Liu, C. Huang, L. Zhang, and T. Zeng, "Transformer for single image super-resolution," 2022. [Online]. Available: https://arxiv.org/abs/2108.11084

[28] X. Zhang, H. Zeng, S. Guo, and L. Zhang, "Efficient long-range attention network for image super-resolution," 2022. [Online]. Available: https://arxiv.org/abs/2203.06697